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A Modern Re-implementation and Empirical Study of MapReduce on Commodity Multiprocessors

作　　者：丁锐

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全文对比

A Modern Re-implementation and Empirical Study of
MapReduce on Commodity Multiprocessors

Rui Ding
Software Engineering Institute East China Normal University Shanghai, China
example@stu.ecnu.edu.cn
Kairui Li
Software Engineering Institute
East China Normal University
Shanghai, China
example@stu.ecnu.edu.cn
Beibei Huang
Software Engineering Institute
East China Normal University
Shanghai, China
example@stu.ecnu.edu.cn
Wenkai Wang
Software Engineering Institute East China Normal University Shanghai, China
71295902070@stu.ecnu.edu.cn

Abstract—MapReduce [1] remains a canonical model for large- scale data processing, yet its original implementation was tied to a specific cluster environment and is largely opaque today. We present x-map-reduce, a from-scratch C++20 re-implementation of the MapReduce programming model that runs on commodity hardware, and report an empirical study of its behaviour.

Our design separates a control plane for metadata from a data plane for bulk transfer, delegates shuffle to worker-to-worker partitions, ships job logic as content-addressed native plugins, and layers a GFS-style block store underneath. Scaling experiments on a single eight-core machine show that increasing the worker count from one to six reduces word-count job time by only 1.

55× on a 10 MB input and 1.46× on a 20 MB input, far from the ideal 6×; on the larger input, going from four to six workers actually makes the job slower. Profiling attributes the gap to two serial stages inside each reduce task—a sequential fetch of all map outputs and a sort-then-tree-insert grouping step—which bound throughput independently of the number of workers.

We quantify the effect of reduce-partition count, which acts as the effective parallelism knob, and discuss what these results imply for small-scale MapReduce deployments.

Index Terms—MapReduce, distributed systems, fault tolerance, performance evaluation, plugin architecture

I. INTRODUCTION

MAPREDUCE, introduced by Dean and Ghemawat [1], provides a simple abstraction for processing large datasets: users supply a map function and a reduce function, and the runtime handles parallelisation, data distribution, and fault recovery.

The model was formulated for a specific environment: hun- dreds of commodity machines in a single datacenter, each with one or two cores, a local disk, and a shared network. Many of its design choices follow from that setting. Computation is pushed to the data because moving bytes across the network was the expensive operation. The coordinator is centralised because the cluster was large enough that the resulting single point of contention was tolerable. Tasks are short and are simply re-run on failure because a recomputation over a local disk is cheap.

This work was completed as a course project.

Modern hardware violates several of these premises. A sin- gle multi-core machine today has more compute than an entire cluster from 2004, with orders of magnitude more memory bandwidth and no network between cores. It is therefore worth asking which parts of the original design still pay off, and which have become the bottleneck. Answering that question requires a system we can instrument, measure, and modify— which existing production implementations, optimised over more than a decade for scale-out deployments, are not well suited to.

This paper reports on x-map-reduce, a from-scratch re- implementation of the MapReduce model in C++20 targeted at commodity multiprocessors, and on what happens when it is measured. We did not set out to build a faster MapReduce; we set out to understand one.

The headline result is negative and, we think, instructive.

Adding workers helps far less than the model suggests: six workers complete a 10 MB word-count job only 1.55× faster than one, and

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on a 20 MB input the sixth pair of workers makes the job measurably slower than four. The reason is not scheduling overhead but two serial stages inside each reduce task, which bound overall throughput independently of how many workers are available. A second finding concerns fault tolerance: a fixed task timeout, calibrated for small inputs, misclassified genuine progress as failure at larger sizes, and the resulting error message named the wrong component. Both findings are properties of the design, not of our implementation effort.

The contributions of this paper are:

A complete re-implementation. We describe x-map-

reduce, an approximately 10k-line C++20 implementa- tion covering scheduling, shuffle, block storage, and fault handling (Sec. III, Sec. IV).

A design that separates control and data planes

and moves shuffle off the coordinator, together with a content-addressed plugin mechanism for job submission (Sec. III).

An empirical characterisation of its behaviour on com-

modity hardware, including scalability, the dominant role of reduce-partition count, and recovery under worker failure (Sec. V), together with the lessons those measure- ments suggest (Sec. VI).

II. RELATED WORK

MapReduce and GFS. The original system [1] and its storage substrate [2] established the model of computation moved to data, with redundant placement and re-execution as the primary fault-tolerance mechanism.

Dataflow successors. Dryad [3], Spark [4], and Flume- Java [5] generalise the two-stage map/shuffle/reduce pipeline to arbitrary dataflow graphs, primarily to avoid materialising intermediate results.

Open-source implementations. Hadoop MapReduce has served as the de-facto reference implementation for over a decade, fixing many of the operational details the original paper leaves implicit—container-based execution, a centralised resource negotiator, and on-disk shuffle.

Speculative execution. The original system guards against stragglers by launching backup tasks near the end of a job. Later work refines the trigger: LATE [6] estimates remaining time from progress rates rather than using a fixed threshold, and Mantri [7] attributes stragglers to causes such as network congestion and re-runs work accordingly. Our implementation uses a static threshold, which Sec. VI identifies as a limitation. Positioning. Our work does not propose a new execution model. It revisits the classic design on hardware that did not exist when the model was defined—commodity multi-core machines rather than racks of single-core disks—and reports where the original design decisions stop paying off.

III. SYSTEM DESIGN

A. Overview

x-map-reduce consists of three roles: a persistent coordi- nator (x-master), persistent workers (x-worker), and a client (xmr-submit). Fig. 1 shows the deployment. Unlike the original design, in which a single master process is created per job and exits when the job finishes, both the coordinator and the workers are long-lived processes. The coordinator serves one job at a time through a per-job coordinator object that is created on submission, drives the job to completion, and then resets; the process itself keeps running and keeps its worker connections and heartbeats alive. This avoids paying connection setup and worker registration costs on every submission, and it makes the failure-detection machinery meaningful—a watchdog is only useful if the things it watches persist across jobs.

A job proceeds in four phases. On submission the client uploads the plugin and provides DFS paths; the coordinator resolves each path to its blocks and creates one map task per block. Map tasks execute on the workers holding the relevant blocks, each partitioning its output into R files, one per reduce partition. Reduce tasks then fetch their partition from every map task’ s host and group by key. Finally the coordinator streams results back to the client and releases the job’ s blocks.

client

xmr-submit

master

Scheduler + Registry

worker A

map output

worker B

submit task

task shuffle P2P

Fig. 1. Deployment and control/data plane separation.

B. Control Plane and Data Plane

The coordinator handles only metadata: submission, worker registration, heartbeats, and task assignment. Bulk data never traverses it. Map outputs stay on the worker that produced them, and the coordinator records only the mapping from map task to host so that reducers know where to fetch. This keeps coordinator message volume a function of task count rather than data size, and it lets the coordinator remain a simple single-threaded reactor.

A consequence is that losing a worker invalidates its map outputs, not just its tasks. The coordinator therefore has to re- run completed map tasks whose output no longer exists, which

we handle by treating such maps as unfinished.

C. Worker to Worker Shuffle

Map output is partitioned at the map side using a hash of the key modulo the number of reduce partitions, so each reducer reads one partition per map task directly from the producing worker’ s data listener. With M map tasks and R reduce partitions the transfer graph is M × R independent point-to- point streams rather than 2M streams funnelled through the coordinator.

D. Content-Addressed Plugins

Job logic ships as a native shared object rather than as code compiled into the workers. Plugins are named by content hash, cached on each worker, and loaded through dlopen; the ABI is a versioned extern "C" entry point returning a job descriptor. Because the identity of a plugin is its hash, workers can cache aggressively and the coordinator never needs to reason about version skew.

E. Block Storage and Locality

Input is stored as fixed-size 1 MiB blocks with replicas placed across workers. One block becomes one map task by default, so the number of map tasks is determined by input size. The coordinator prefers to assign a map task to a worker that already holds the block, falling back to any idle worker when no replica is free, which yields data locality without a separate scheduler.

F. Design Trade-offs

Three of these decisions carry costs that only became visible once the system was measured, and we state them explicitly

rather than presenting the design as unproblematic.

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Peer-to-peer shuffle is not free. Keeping bulk data off the coordinator makes the coordinator scale, but it makes each reducer responsible for gathering its own input. Because one

TABLE I

IMPLEMENTATION SIZE BY MODULE (LINES OF C++)

Module	Lines
protocol	2 162
tests	1 904
master	1 836
dfs	1 208
runtime	1 098
worker	950
client	658
mapreduce	334
server	265
jobs	69
total	10 484

block is one map task, a reducer must contact as many peers as there are blocks. Making the block size smaller improves parallelism but increases this fan-in; making it larger reduces fan-in but coarsens scheduling.

Our 1 MiB choice, inherited from the block-storage layer, is a compromise that we did not tune, and Sec. V shows the consequence.

Content addressing assumes compatible machines. Ship- ping native code works because every worker runs the same binary interface on the same architecture. Across heteroge- neous hardware a plugin cache keyed by content hash would be incorrect, and the mechanism would have to fall back to source distribution or bytecode.

This is an acceptable assumption for a single-machine deployment and a poor one for a mixed cluster. A centralised coordinator remains a single point of failure. Replicating it requires persistent job state, which the current implementation does not keep; recovering a coordina- tor after restart would therefore lose in-flight jobs even though worker data survives.

Our fault experiments consequently cover worker loss only.

IV. IMPLEMENTATION

The implementation is approximately 10 000 lines of C++20. Table I gives the per-module breakdown.

A. Event-Driven Runtime

Each process runs an epoll-based reactor. Network I/O is handled by a small number of I/O threads, while task execution—map and reduce functions—is offloaded to a worker thread pool so that a long computation never blocks the event loop. The coordinator runs a fixed-interval tick, mod- elled on Pregel’ s super-steps, that drives scheduling, heartbeat expiry, and task timeouts.

B. Wire Protocol

All control messages share one binary framing: a 16-byte big-endian header carrying magic (XMRP), version, type, flags, a request identifier, and a payload length capped at 64 MiB, followed by a variable-length body. Strings and blobs are length-prefixed rather than delimiter-separated, so the codec is binary-safe. Framing is the only thing the transport layer

TABLE II

JOB DURATION (MEDIAN OF 3 RUNS) AND SPEED-UP RELATIVE TO ONE WORKER

	10 MB		20 MB	
Workers	s	Speed-up	s	Speed-up
1	18.67	1.00×	37.86	1.00×
2	15.53	1.20×	28.58	1.32×
4	13.03	1.43×	25.48	1.49×
6	12.01	1.55×	25.88	1.46×
ideal		W×		W×

knows; message semantics live in a separate module, and ver- sion mismatch is resolved by a HELLO/HELLO ACK exchange before any other message is accepted.

C. Fault Tolerance Mechanisms

Three mechanisms operate together. A heartbeat watchdog marks a worker lost after 6 s of silence and requeues whatever it held. A per-task watchdog re-dispatches a task that has not reported completion within 30 s, even if its worker is still alive. Speculative execution duplicates a task when it has run longer than max(3 s, 2× fastest completed attempt) and a spare worker is idle; the first attempt to finish wins and the loser is cancelled.

Retries are idempotent by construction: each attempt carries an identifier, and a late result from a superseded attempt is discarded rather than allowed to overwrite the current one. After four attempts a task is declared failed and the job aborts. As Sec. V shows, this fixed timeout is the component most sensitive to workload scale.

V. EVALUATION

A. Experimental Setup

All experiments run on a single machine with an 8-core Intel Xeon Platinum 8272CL (2.60 GHz, one thread per core) and 7.5 GB of RAM running CentOS Stream 9. The coordinator and all workers are separate processes on this machine com- municating over the loopback interface; we cap concurrency at six workers to leave headroom for co-resident services.

The workload is word count over synthetic text. Input is generated to a fixed byte size with a vocabulary of 2 000 distinct tokens and five tokens per line. Because the input is partitioned into 1 MiB blocks and each block becomes one map task, a 10 MB job comprises ten map tasks. Input files are ingested into the block store before each measurement, and job duration is measured from submission until the coordinator acknowledges the result, so it includes scheduling, mapping, shuffle, and reduce.

Each configuration is measured three times; we report the median.

B. Scalability

Fig. 2 plots job duration against worker count for 10 MB and 20 MB inputs, and Table II gives the medians of three runs per configuration.

1 2 4 6

0

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10 20 30 40
Number of workers
Jo b
du ra tio n
(s)
20 MB
10 MB
ideal (20 MB)

Fig. 2. Job duration versus worker count. Both curves flatten well before the available core count, and the larger input turns upward at six workers.

Adding workers yields strongly diminishing returns on both inputs. For 10 MB the first extra worker buys 1.20×, but the last two workers together add only 8%; total duration falls from 18.67 s to 12.01 s over a six-fold increase in workers, a speed-up of 1.55×.

The 20 MB curve is more informative. It tracks closer to linear up to four workers (1.49×) and then regresses: six workers take 25.88 s against 25.48 s at four. The job gets slightly slower when given more resources. This is the signature of a bottleneck that is not shared and therefore cannot be amortised: past four workers, additional workers add contention and scheduling overhead without shortening the critical path.

C. Methodology and Threats to Validity

Before interpreting the curve we address factors that could produce it artificially, since a flat scaling curve is easy to obtain for the wrong reasons.

Co-resident services. The machine also runs unrelated containerised services. We cap the worker count at six on eight cores and leave the rest for those services. This means absolute timings are slightly pessimistic, but it does not explain the shape of the curve: with six workers and two spare cores, neither core oversubscription nor memory exhaustion occurred in these runs.

Measurements include fixed costs. Job duration is mea- sured from submission to result acknowledgement, so it in- cludes process start-up, worker registration, and plugin distri- bution. These are per-job rather than per-worker costs, and they inflate all points roughly equally—which if anything would flatten the curve. To check, we measured a 10 MB input ingestion alone at 1.6–2.4 s, a small fraction of the 12–20 s job time, confirming that fixed I/O overhead is not dominant. Run-to-run variance. Each configuration is measured three times. Observed spread is within about 10% of the median, well below the differences between configurations, so the ordering of the medians is stable across repetitions.

Single host. All workers share one machine and commu- nicate over loopback. This removes network cost from the shuffle entirely, and is therefore the most favourable setting

20 22 23 0 5
10
15
20 19.2
12.3 13.0
Number of reduce partitions R
Jo
b
du
ra
tio
n
(s
)

Fig. 3. Effect of reduce-partition count at four workers, 10 MB input. Four partitions is the sweet spot; beyond it, duration increases again.

for scaling in which shuffle is the constraint. Any bottleneck we observe here can only be worse when workers are spread across physical hosts. Conversely, the experiment cannot speak to network effects, which is a limitation we return to in Sec. VI.

D. Why Scaling Stalls: The Reduce Bottleneck

The flat curve is not a scheduling artefact. Profiling the implementation identifies two serial stages that bound overall throughput regardless of how many workers are available. First, a reduce task fetches the output of every map task into a single in-memory vector before it begins reducing (runtime/src/task.cpp). The number of map tasks grows linearly with input size, so the per-reducer working set

grows with the job, and this fetch loop is sequential.

Second, grouping sorts the entire intermediate dataset and then inserts every element into an ordered map (mapreduce/src/mapreduce.cpp). The sort already es- tablishes key order, yet an additional tree insertion per element follows, making this stage O(n log n) with a large constant. Both stages execute inside a single reduce task, so they do not overlap across workers. Consequently overall time is dominated by the slowest reducer rather than by the aggregate map throughput.

E. Effect of Reduce Partitions

Since the reduce stage is serial per task, the number of reduce partitions is the parallelism knob. In a preliminary sweep over a 10 MB input with four workers, durations were

19.21 s, 12.28 s, and 13.04 s for one, four, and eight reducers respectively (Fig. 3). Moving from one to four partitions cuts duration by roughly a third, while increasing further to eight yields no benefit—consistent with the available worker count,

beyond which extra partitions only add overhead.

The shape is explained by the two competing effects. Too few partitions concentrates all intermediate data into a single serial reduce task. Too many partitions creates more tasks than there are workers, so partitions queue behind one another while each still pays the fixed cost of fetching from all map tasks. The optimum therefore tracks the worker count rather than the data size—which is itself a symptom of the underlying

TABLE III
FAULT RECOVERY: ONE WORKER KILLED MID-JOB (4 WORKERS, 10 MB) Kill after Duration (s) Recovered Output keys
1 s 12.42 ✓ 2 000
2 s 12.57 ✓ 2 000

3 s 13.29 ✓ 2 000

problem, because a well-partitioned shuffle need not care how many workers run the reduce phase.

F. Interaction Between Memory Pressure and Task Timeouts A second, less expected failure mode appeared at larger input sizes and is worth reporting because its diagnosis is counter-intuitive.

On a 50 MB input with four workers, jobs failed consis- tently, always after roughly 30 seconds, with the error map task 0 failed: attempts exhausted. Two obser- vations seemed contradictory. First, an isolated 1 MiB map task completes in about 1.26 s, far below any plausible timeout. Second, sampling resident memory during the run shows aggregate worker RSS climbing monotonically from 0.2GB at 2 s to 2.65GB at 30 s before falling back:

t (s) 2 10 20 30 34

RSS (GB) 0.22 0.72 1.34 2.65 0.96

The memory growth originates in reduce, not map: each reduce task pulls the output of all fifty map tasks into memory at once before sorting. Under this pressure the worker's event loop slows, map tasks overrun the coordinator's hard-coded 30 s kTaskTimeout, and the coordinator re-dispatches them. After four attempts (maxAttempts = 4) the job is declared failed. The reported error blames map task 0, but the root cause lies entirely in the reduce path.

To confirm, we raised kTaskTimeout from 30 s to 600 s, recompiled, and re-ran the identical 50 MB job. It completed successfully in 56.3 s with correct output (2 000 distinct keys). The system was always capable of the work; the timeout policy, calibrated for small inputs, was misclassifying progress as failure. Increasing the number of reduce partitions did not help—the job still failed at 30.1 s with sixteen partitions— because partition count does not reduce the total intermediate data each worker must buffer.

G. Fault Recovery

We inject a fault by killing one worker with SIGKILL at a fixed delay after job submission, leaving three of the four workers alive, and measure whether the job still completes. Table III reports the results over a 10 MB input; the fault-free baseline for the same configuration is 12.46–13.42 s. All three runs completed with correct output, and the overhead relative to the fault-free baseline is negligible (0.0%, 0.9%, and 6.6%). The recovery path is therefore cheap at this scale: the coordinator detects the loss through the heartbeat

mechanism, re-dispatches the affected tasks to surviving work- ers, and the job proceeds. The later the fault is injected, the more work has already completed, so the marginal cost grows slightly.

We note that this experiment exercises worker loss, not coordinator loss. The coordinator remains a single point of failure in the current implementation, as discussed in Sec. VI.

VI. LESSONS LEARNED

Building and measuring the system surfaced several ob- servations that we believe transfer beyond this particular implementation.

Control/data plane separation pays off structurally, not necessarily in wall-clock time. Moving shuffle off the coordinator was motivated by the original design's single-point bottleneck. It succeeds in keeping the coordinator's message volume small and independent of job size: the coordinator only exchanges metadata, never bulk bytes. Yet our measurements show that end-to-end time is now set by per-reduce-task work rather than by coordinator throughput. Removing one bottleneck exposed the next, and the structural fix alone did not change the scaling curve.

Fault-tolerance parameters are workload-dependent and can silently convert progress into failure. The hardest bug we encountered was not a crash but a misclassification. Reduce tasks buffering the output of all map tasks drove worker memory to 2.65GB, slowing the event loop enough that map tasks exceeded a hard-coded 30 s timeout. The coordinator dutifully retried four times and reported map task 0 failed, naming the wrong component. Raising only the timeout made the identical job succeed in 56 s. A fixed timeout cannot be correct across input sizes; progress reporting, which the system already collects for speculation, is the more appropriate signal for deciding whether a task is alive.

Small implementation details dominate at this scale. The grouping step sorts the intermediate data and then inserts every element into an ordered map. The sort has already established key order, so the tree insertion is redundant work; a linear pass over sorted input would suffice. At large scale this constant factor would be lost in the noise. At the scale we can measure,

it accounts for most of the reduce time.

What we would do differently. Persistent metadata for the block namespace, coordinator replication, and multi-job concurrency all remain unimplemented, and the coordinator is still a single point of failure. Our fault experiment measures worker loss only. We would prioritise making the coordinator's timeout policy adaptive, and replacing the sort-then-tree group- ing with a streaming merge, before adding further features.

VII. CONCLUSION

We presented x-map-reduce, a from-scratch C++20 re- implementation of MapReduce on commodity hardware, and characterised its behaviour empirically. The design separates control and data planes, performs shuffle directly between workers, and delivers job logic as content-addressed native plugins over a GFS-style block store.

Our measurements show that the system scales only weakly: six workers improve a 10 MB word-count job by 1.55× and a 20 MB job by 1.46× over a single worker, not the ideal 6×; on the larger input the curve turns upward beyond four workers. We traced this to two serial stages within each reduce task—a sequential fetch of every map output and a sort-then-tree-insert grouping step—which bound throughput independently of worker count.

Separately, we found that reduce-stage memory growth can push map tasks past a fixed timeout, producing a failure report that misattributes the cause; relaxing only the timeout makes the same job succeed. These results suggest that for small-scale, single-machine MapReduce deployments, the reduce path and its resource behaviour deserve more attention than the number of workers.

Extending the study to multiple physical hosts, adding multi- job concurrency, and replacing the grouping implementation are the natural next steps.

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